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Research paper

Physics-informed neural networks for 3D transient thermal simulation of MIG welding

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ABSTRACT

Accurate and fast prediction of transient temperature fields in MIG/MAG welding is essential for process qualification and weld integrity, yet high fidelity mesh-based simulations can become computationally expensive for fully transient 3D problems with moving heat sources. To circumvent these computational bottlenecks, this study delineates an advanced simulation architecture founded upon Physics-Informed Neural Networks (PINNs). This paradigm shifts the solution strategy from mesh-based integration to physics-constrained optimization, ensuring thermodynamic consistency while significantly accelerating time-to-solution. The proposed architecture directly assimilates the partial differential equation governing three-dimensional transient heat conduction, incorporating a dynamic Gaussian volumetric energy source, into the network's objective function. Consequently, the training process minimizes the residuals of the governing physics, constraining the predicted manifold to satisfy Fourier's law of conduction without the necessity for spatial discretization. Empirical validation against previously published bead-on-plate experiments on 6061-T6 aluminum alloy substrates confirms the model's fidelity. The framework successfully reproduces complex transient thermal profiles, yielding a minimized l_2 error metric of 2.28 K, demonstrating robust agreement with thermocouple telemetry. Furthermore, a systematic hyperparameter study was conducted, as PINN performance is highly sensitive to parameter choices. Finally, this study identified an optimal configuration that balances predictive accuracy with computational cost. Notably, the optimized hyperparameters facilitate model convergence in approximately 14 min utilizing a single consumer-grade graphics processing unit. This efficiency establishes the PINN framework as a viable engine for real-time digital twinning and rapid iterative design in high-throughput manufacturing environments.

1. Introduction

Metal inert gas (MIG) and metal active gas (MAG) welding are critical manufacturing processes, favored for their high feeding rates and productivity [1–3], even when compared to advanced variants of arc [4] and laser processes [5]. However, this productivity introduces complex thermo-mechanical phenomena. The rapid heating and cooling cycles induce residual stresses and microstructural transformations that define the component's service life. Consequently, predicting the spatiotemporal thermal history of a weldment is a critical requirement for process certification and quality assurance.

Historically, simulating these thermal fields has relied on mesh-based numerical methods, such as the finite element method (FEM) and finite volume method (FVM). While effective for analyzing weld pool dynamics [6,7], these approaches are constrained by spatial discretization. Resolving the steep temperature gradients of a moving arc requires high mesh density, resulting in systems with millions of degrees of freedom. This renders transient 3D simulations computationally prohibitive for

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real-time applications or digital twin development, particularly given the non-linear scaling of computational costs with the fine time steps required to capture rapid heat source dynamics.

To address these limitations, scientific machine learning (SciML) offers a paradigm shift: using neural networks not merely as data interpolators, but as functional approximators of differential equations. Physics-informed neural networks (PINNs) exemplify this approach [8]. Unlike traditional deep learning, which relies solely on labeled data, PINNs embed governing physical laws, such as energy conservation–directly into the loss function. By using automatic differentiation to compute derivatives without truncation errors, PINNs provide a mesh-free solution advantageous for high-dimensional domains and moving boundaries [9].

The application of machine learning to welding and additive manufacturing (AM) has grown rapidly. Recent studies have utilized standard artificial neural networks and hybrid models as surrogates to reduce reliance on experimental trials. For instance, Rawa et al. [10] optimized laser welding parameters for dissimilar steels using neural networks, while [11] integrated Gaussian processes to predict deep-penetration laser welding thresholds. In the domain of solid-phase processing, King et al. [12] developed neural networks for temperature prediction in shear assisted process extrusion, and [13] achieved rapid temperature profiling in laser powder-bed fusion using feed-forward networks trained on FEM data. These works, along with hybrid frameworks for direct metal deposition [14], demonstrate the utility of machine learning in optimizing thermal behavior with high computational efficiency.

Within the specific niche of physically guided learning, Li et al. [15] introduced a PINN framework for 3D temperature fields in laser metal deposition, incorporating temperature-dependent properties to achieve accuracy comparable to FEM but with superior efficiency. This trend is further supported by [16], who reviewed physics informed machine learning in metal additive manufacturing, highlighting its potential for real-time responsiveness. However, a distinct gap remains in the literature: while PINNs have been applied to laser-based [17,18] and additive [19,20] processes, their application to the specific physics of MIG welding remains not well studied.

MIG welding differs fundamentally from the processes currently addressed in PINN literature. Unlike layer-wise additive manufacturing, which involves rapid, cyclic heating of small melt pools, MIG welding involves continuous, high-energy input into a massive substrate. This generates long-range heat diffusion and stiff temperature gradients without the thermal “reset" inherent in multi-layer AM. Furthermore, while laser welding is typically modeled with surface fluxes (Neumann boundary conditions), MIG welding relies on a consumable electrode best represented by volumetric heat sources (e.g., Gaussian or Goldak distributions). This presents a non-trivial mathematical challenge for PINNs: the network must resolve internal heat generation terms that move continuously through the 3D domain. Standard collocation methods often struggle to track such moving sources, leading to convergence toward trivial solutions where PDE residuals are minimized via negligible temperature variation.

Addressing these challenges aligns with the growing need for sustainable manufacturing. As highlighted by [21], sustainability in welding requires minimizing energy-intensive experimental trials. By embedding physical constraints, PINNs can provide accurate predictions with minimal experimental validation, reducing material waste and energy consumption.

In this sense, this work presents a PINN-based framework for the full-scale 3D thermal simulation of MIG welding, specifically addressing the challenge of reconstructing transient thermal fields from sparse experimental data. Distinct from existing studies that predominantly validate against dense synthetic data generated by finite element analysis [22], effectively performing model distillation, this approach bridges the ‘sim-to-real’ gap by assimilating sparse signals from physical thermocouples. By embedding the transient heat conduction equation and a moving volumetric Gaussian heat source directly into the training framework, the proposed model captures the continuous propagation of high-intensity heat throughout a large domain. The validation demonstrates that the framework can accurately reconstruct full 3D fields from limited sensor inputs, achieving computational efficiency comparable to surrogates used in other manufacturing sectors without relying on the perfect field knowledge assumed in synthetic data studies.

2. Methods

2.1. Physics-informed neural networks (PINNs)

PINNs have gained significant traction in the literature for embedding physical laws directly into neural network frameworks. This approach enables the resolution of initial-boundary value problems governed by partial differential equations (PDEs) ([8]). Within the PINN framework, the governing equations, initial conditions, and boundary conditions are integrated into the loss function, driving the model to approximate solutions that align with these constraints.

A general initial-boundary value problem involving a PDE can be described over a spatial domain Ω and a temporal domain Γ, as follows:

$$\mathcal{D}[u(\boldsymbol{x},t)] = f(\boldsymbol{x},t), \quad \boldsymbol{x}\in\Omega,\ t\in\Gamma, \tag{1}$$

$$u(\boldsymbol{x},0) = u_0(\boldsymbol{x}), \quad \boldsymbol{x}\in\Omega, \tag{2}$$

$$u(\boldsymbol{x},t) = g(\boldsymbol{x},t), \quad \boldsymbol{x}\in\partial\Omega,\ t\in\Gamma, \tag{3}$$

where $\mathcal{D}$ is the differential operator, $u(\boldsymbol{x},t)$ is the unknown function, $f(\boldsymbol{x},t)$ represents the source term, $u_0(\boldsymbol{x})$ specifies the initial condition, and $g(\boldsymbol{x},t)$ defines the boundary conditions on $\partial\Omega$.

PINNs estimate the solution $\hat{u}\approx u$ by minimizing the residuals of the PDE and associated conditions. These residuals contribute to three main loss terms: $\mathcal{L}_{\text{pde}}$, $\mathcal{L}_{\text{ic}}$, and $\mathcal{L}_{\text{bc}}$, which correspond, respectively, to the accuracy of the solution in the domain, at the initial conditions, and along the boundaries:

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_{\text{pde}}}\sum_{i=1}^{N_{\text{pde}}}\left|\mathcal{D}\left[\hat{u}\left(\boldsymbol{x}^i_{\text{pde}},t^i_{\text{pde}};\theta\right)\right] - f\left(\boldsymbol{x}^i_{\text{pde}},t^i_{\text{pde}}\right)\right|^2, \tag{4}$$

$$\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{N_{\text{ic}}}\sum_{i=1}^{N_{\text{ic}}}\left|\hat{u}\left(\boldsymbol{x}^i_{\text{ic}},0;\theta\right) - u_0\left(\boldsymbol{x}^i_{\text{ic}}\right)\right|^2, \tag{5}$$

$$\mathcal{L}_{\text{bc}}(\theta) = \frac{1}{N_{\text{bc}}}\sum_{i=1}^{N_{\text{bc}}}\left|\hat{u}\left(\boldsymbol{x}^i_{\text{bc}},t^i_{\text{bc}};\theta\right) - g\left(\boldsymbol{x}^i_{\text{bc}},t^i_{\text{bc}}\right)\right|^2, \tag{6}$$

where θ are the neural network's parameters, and N_{pde}, N_{ic}, and N_{bc} denote the number of collocation points used for the domain, initial, and boundary conditions, respectively.

The overall loss function $\mathcal{L}$ combines these components, typically as a weighted sum:

$$\mathcal{L}(\theta) = \lambda_{\text{pde}}\mathcal{L}_{\text{pde}}(\theta) + \lambda_{\text{ic}}\mathcal{L}_{\text{ic}}(\theta) + \lambda_{\text{bc}}\mathcal{L}_{\text{bc}}(\theta) \tag{7}$$

where λ_{pde}, λ_{ic}, and λ_{bc} are weighting coefficients that serve to balance the disparate magnitudes of the loss terms. While these hyperparameters can be updated dynamically using adaptive algorithms (e.g., [23–25]), the present study employs a static weighting strategy to isolate the baseline performance of the architecture. We explicitly set all coefficients to unity ($\lambda_{\text{pde}} = \lambda_{\text{ic}} = \lambda_{\text{bc}} = 1$) and maintain them fixed throughout the entire optimization process, without the use of annealing schedules or gradient balancing techniques. Training the PINN involves minimizing $\mathcal{L}(\theta)$ to optimize the neural network parameters θ, ensuring that the predicted solution $\hat{u}(\boldsymbol{x},t;\theta)$ satisfies the governing PDE and associated conditions. This optimization is expressed as:

$$\theta^* = \arg\min_{\theta}\ \mathcal{L}(\theta) \tag{8}$$

where θ^* denotes the optimal parameter set that provides the approximate solution $\hat{u}\approx u$.

Gradient-based optimization methods, such as [26], are commonly used to solve this minimization problem. During training, the gradients

of the loss function with respect to θ are calculated using automatic differentiation, enabling efficient and accurate updates to the network parameters.

2.1.1. Governing equations

In MIG/MAG welding, heat transfer is governed by the transient three-dimensional heat conduction equation, incorporating a moving heat source that represents the welding arch. The governing equation can be written as:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q(\boldsymbol{x}, t), \quad \boldsymbol{x} \in \Omega \subset \mathbb{R}^3, \ t \in [0, t_f] \tag{9}$$

where $\boldsymbol{x} = (x, y, z)^\top$ is the position vector, Ω is the spatial domain, T is the temperature field, ρ is the material density, c_p is the specific heat capacity, k is the thermal conductivity, and $Q(\boldsymbol{x}, t)$ represents the heat input from the welding source per unit volume. This equation is embedded as a strong constraint within the PINN framework via a physics-based loss term, $\mathcal{L}_{\text{pde}}$, ensuring that predictions adhere to the governing laws of heat conduction.

The heat source $Q(\boldsymbol{x}, t)$ is typically modeled as a Gaussian distribution centered at the instantaneous position of the welding torch:

$$Q(\boldsymbol{x}, t) = Q_0 \exp\left(-\frac{r^2}{r_0^2}\right) \tag{10}$$

where Q_0 denotes the maximum volumetric heat generation rate (W/m^3) at the center of the Gaussian, $r = \sqrt{(x - x_s)^2 + (y - y_s)^2 + (z - z_s)^2}$ is the radial distance from the moving focus (x_s, y_s, z_s), and r_0 is a characteristic radius of the Gaussian distribution.

Specific boundary conditions (e.g., Neumann, Dirichlet, Robin, etc) are needed to ensure an accurate definition of heat flow and temperature distribution in the domain, consistent with the Gaussian heat source model and the physical constraints of the MIG/MAG welding process. For the present study, the above-mentioned volumetric Gaussian heat source was used. Details about the boundary conditions will be presented in Section 4.

2.1.2. Thermal modeling with PINNs

The proposed framework for modeling MIG/MAG welding employs PINNs combined with physics-based constraints and optionally augmented by data-driven loss functions. This methodology integrates fundamental physical laws with experimental data to predict the thermal behavior during the welding process, capturing the spatiotemporal temperature distributions characteristic of the welding processes.

To effectively capture the contributions of different physical equations within the loss function, the residual terms of Eqs. (4)–(6) are reformulated as follows, where $T(\boldsymbol{x}, t)$ represents the model's predicted temperature output.

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_{\text{pde}}} \sum_{i=1}^{N_{\text{pde}}} \left| \rho c_p \frac{\partial T}{\partial t}\bigg|_{\left(\boldsymbol{x}^i_{\text{pde}}, t^i_{\text{pde}}\right)} - \nabla \cdot (k \nabla T)\bigg|_{\left(\boldsymbol{x}^i_{\text{pde}}, t^i_{\text{pde}}\right)} - Q\left(\boldsymbol{x}^i_{\text{pde}}, t^i_{\text{pde}}\right) \right|^2, \tag{11}$$

$$\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{N_{\text{ic}}} \sum_{i=1}^{N_{\text{ic}}} \left| T\left(\boldsymbol{x}^i_{\text{ic}}, 0; \theta\right) - t_0\left(\boldsymbol{x}^i_{\text{ic}}\right) \right|^2, \tag{12}$$

$$\mathcal{L}_{\text{bc}}(\theta) = \frac{1}{N_{\text{bc}}} \sum_{i=1}^{N_{\text{bc}}} \left| T\left(\boldsymbol{x}^i_{\text{bc}}, t^i_{\text{bc}}; \theta\right) - g\left(\boldsymbol{x}^i_{\text{bc}}, t^i_{\text{bc}}\right) \right|^2. \tag{13}$$

To enhance the model's accuracy, a data-driven loss term, $\mathcal{L}_{\text{data}}$, can be incorporated alongside the physics-based loss. This term uses experimental temperature data to refine predictions in regions where empirical insights are available:

$$\mathcal{L}_{\text{data}}(\theta) = \frac{1}{N_{\text{obs}}} \sum_{i=1}^{N_{\text{obs}}} \left\| T(\boldsymbol{x}^i_{\text{obs}}, t^i_{\text{obs}}) - T^i_{\text{obs}} \right\|^2 \tag{14}$$

where N_{obs} denotes the number of observation points, and T_{obs} represents the measured temperatures.

Accurately solving the heat equation also requires enforcing boundary and initial conditions, which are incorporated as additional loss terms:

- Initial condition loss ($\mathcal{L}_{\text{ic}}$) ensures that the model's predicted temperature field aligns with the specified initial temperature distribution.
- Boundary condition loss ($\mathcal{L}_{\text{bc}}$) enforces correct heat fluxes or temperature values at the domain boundaries.

The total loss function combines these terms, balancing the contributions of physics-based and data-driven components:

$$\mathcal{L} = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}} + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}} \tag{15}$$

where $\lambda_{\text{pde}}, \lambda_{\text{ic}}, \lambda_{\text{bc}}, \lambda_{\text{data}}$ are weighting coefficients that regulate the influence of each loss term on the overall optimization process. By embedding physical principles and incorporating experimental data, the framework effectively captures the complex heat transfer phenomena that are central to the welding process.

Fig. 1 illustrates the architecture of the PINN model. The PINN is trained by minimizing a weighted sum of mean-squared errors. The inputs consist of spatial coordinates, $\boldsymbol{x}$, and the temporal feature t, which align with the variables used in the governing partial differential equations. The output of the model is the predicted temperature field T. Automatic differentiation [27] is used to compute the first- and second-order derivatives of T with respect to $\boldsymbol{x}$ and t. These derivatives are then incorporated into the loss function to evaluate the physical losses during backpropagation. Material properties, process variables, and welding parameters are treated as known physical constants that enter the governing equations and boundary conditions; they are not hyperparameters of the PINN. The model hyperparameters refer to the network architecture and training settings (e.g., number of layers/neurons, activation function, learning rate, and loss weights).

In real-world manufacturing, factors such as work piece size, deposition time, and melt pool temperature vary significantly across physical scales. For instance, spatial features are often in the millimeter range, while the melt pool peak temperature can exceed 1000 K. Directly using the raw physical inputs for the model is impractical, as the magnitude disparity between spatial gradients and temperature gradients can destabilize training and hinder convergence.

To address this, the network input is explicitly formulated as a tensor $\mathbf{X} \in \mathbb{R}^{N \times 4}$, rather than a discretized grid. This input tensor represents a batch of spatiotemporal coordinates $\mathbf{X} = \{(\boldsymbol{x}_i, t_i)\}_{i=1}^{N}$, where N is the batch size combining both collocation and experimental data points. Before passing this tensor to the network, a scaling method is applied to map these values to a normalized range of $[0, 1]$. This approach adjusts the spatial coordinates, time steps, and temperature by multiplying them with appropriate scaling coefficients k. After rescaling from $(\boldsymbol{x}, t) \to T$ to the dimensionless space $(\boldsymbol{x}', t') \to T'$, the original residual term from Eq. (11) is reformulated as:

$$\boldsymbol{x}' = k_x \boldsymbol{x}, \tag{16}$$

$$t' = k_t t, \tag{17}$$

$$T' = k_T T, \tag{18}$$

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_{\text{pde}}} \sum_{i=1}^{N_{\text{pde}}} \left| \frac{k_T}{k_t} \rho c_p \frac{\partial T}{\partial t'}\bigg|_{\left(\boldsymbol{x'}^i_{\text{pde}}, t'^i_{\text{pde}}\right)} - \frac{k_T}{k_x^2} \nabla \cdot \left(k \nabla T'\right)\bigg|_{\left(\boldsymbol{x'}^i_{\text{pde}}, t'^i_{\text{pde}}\right)} - Q\left(\boldsymbol{x'}^i_{\text{pde}}, t'^i_{\text{pde}}\right) \right|^2. \tag{19}$$

The data processing pipeline begins by providing raw time-series from four thermocouples, where temperatures are converted to Kelvin and temporally aligned to $t = 0$ without applying downsampling or outlier removal. Supervision pairs are formed by mapping spatiotemporal coordinates to measured temperatures, with all probes located at the geometric centerline but positioned at varying depths ($z = 0.026$–0.035

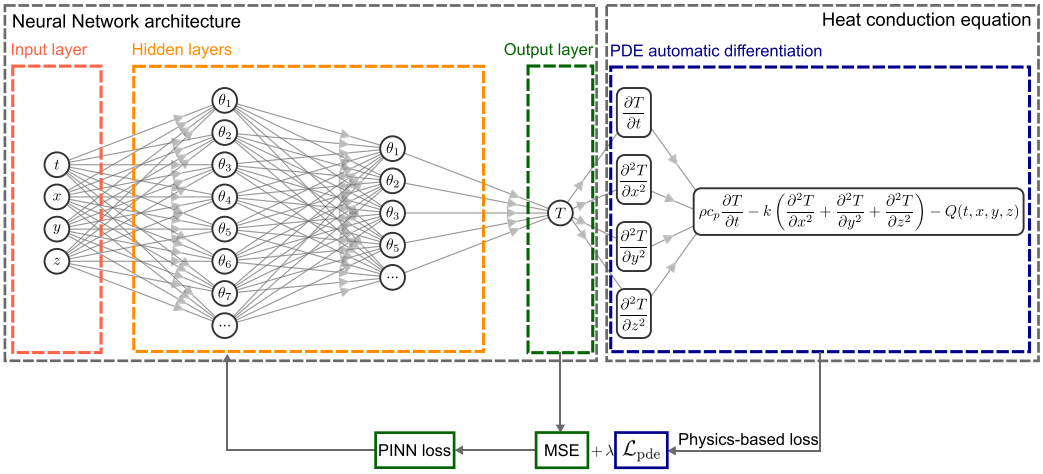


Fig. 1. Physics informed neural network model architecture.

m). The modeling framework integrates this data by defining a neural network architecture $\mathcal{N}_\theta$ which approximates the temperature $T(\boldsymbol{x}, t)$. Rather than employing a traditional train-test split, the entire experimental dataset acts as sparse ground-truth supervision alongside the physical constraints–including the partial differential equation, the heat source, and boundary/initial conditions. A composite loss function is defined to guide the training process, comprising: $\mathcal{L}_{\text{pde}}$, which enforces the heat equation by penalizing deviations of the PDE residual at randomly sampled collocation points; $\mathcal{L}_{\text{bc}}$ and $\mathcal{L}_{\text{ic}}$, which incorporate boundary and initial conditions; and $\mathcal{L}_{\text{data}}$, which ensures agreement with the experimental measurements. The neural network is trained by minimizing the total loss $\mathcal{L}$ using gradient-based optimization methods, with gradients computed via automatic differentiation. Upon completion, the trained network $\mathcal{N}_\theta$ serves as a surrogate model for the temperature distribution $T(\boldsymbol{x}, t)$, satisfying both physics-based and data-driven constraints. An outline of the pseudocode used in this work is shown in Algorithm 1.

Algorithm 1 PINN pseudocode for a 3D heat equation with a moving heat source.

Input: Neural network architecture $\mathcal{N}_\theta$ with trainable parameters θ
Given:
- Physical domain: $(\boldsymbol{x}, t)$
- Partial differential equation: $\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q(\boldsymbol{x}, t)$
- Heat source $Q(\boldsymbol{x}, t)$: Moving Gaussian
- Experimental data: $\{x_i, y_i, z_i, t_i, T_{\text{exp},i}\}_{i=1}^{N_{\text{exp}}}$
- Boundary and initial conditions

Normalize all input data (x, y, z, t, T)
Define the total loss function:

$$\mathcal{L} = \mathcal{L}_{\text{pde}} + \mathcal{L}_{\text{bc}} + \mathcal{L}_{\text{ic}} + \mathcal{L}_{\text{data}} \tag{20}$$

Initialize $\theta \leftarrow$ Random Initialization
while not converged **do**
    Compute predictions u_θ using $\mathcal{N}_\theta(\boldsymbol{x}, t)$
    Compute gradients $\frac{\partial T_\theta}{\partial t}$, $\frac{\partial^2 T_\theta}{\partial x^2}$, $\frac{\partial^2 T_\theta}{\partial y^2}$, and $\frac{\partial^2 T_\theta}{\partial z^2}$ using automatic differentiation
    Evaluate $\mathcal{L}_{\text{pde}}$, $\mathcal{L}_{\text{bc}}$, $\mathcal{L}_{\text{ic}}$, and $\mathcal{L}_{\text{data}}$
    Compute total loss $\mathcal{L}$
    Update parameters $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ using a gradient-based optimizer
end while
Output: Trained network $\mathcal{N}_\theta$ approximating $T(\boldsymbol{x}, t)$

Table 1
Welding parameters used by [28].

Parameter	Value
Voltage (V)	23.1
Current (A)	167
Travel speed (mm/s)	16.5
Wire feed speed (mm/s)	124.9
Wire diameter (mm)	1.2
Gas flow (L/min)	14.16
CTWD[1] (mm)	18

[1]Contact tip-to-workpiece distance.

In order to facilitate the training of the PINN and improve its convergence properties, the governing equations are rendered dimensionless. By solving the problem in this dimensionless space, the PINN training is more stable and general, and the final results can be rescaled back to the original physical units once the solution is obtained.

3. Experimental procedure

In order to validate the PINN modeling and perform additional tests, experimental data of [28] were used. In this experiment, the temperature evolution is investigated by four thermocouples located at the mid-length of the weld line. Fig. 2 show the sample dimensions, weld line and thermocouple locations.

Champagne and Pham [28] experimental test was conducted using a DX-100 Yaskawa robot and a Fronius TransPuls Synergic 4000 power source, performing bead-on-plate welds with the MIG process and Argon shielding gas. The samples, made of cold-rolled 6061-T6 aluminum (150 mm × 9.5 mm × 38 mm), were mounted vertically to minimize arc radiation effects on the thermocouples. These thermocouples were spot-welded along the sides of the samples at varying distances from the weld line to record temperatures near the weld bead. Four 0.5 mm diameter type-K thermocouples, connected to a Graphtec GL220 data acquisition module, recorded temperature data at a rate of 10 readings per second. The welding parameters and thermal properties of 6061-T6 alloy are given in Tables 1 and 2, respectively and were used as inputs for the PINN. For the sake of simplicity, the heat process will be modeled using a moving Gaussian heat source described in Section 2.1.1, and only the heat transfer process will be evaluated in the present investigation.

4. Numerical procedure and framework

An in-house code, named WelPhys AI (an acronym for Welding Physics Artificial Intelligence), was developed specifically for the

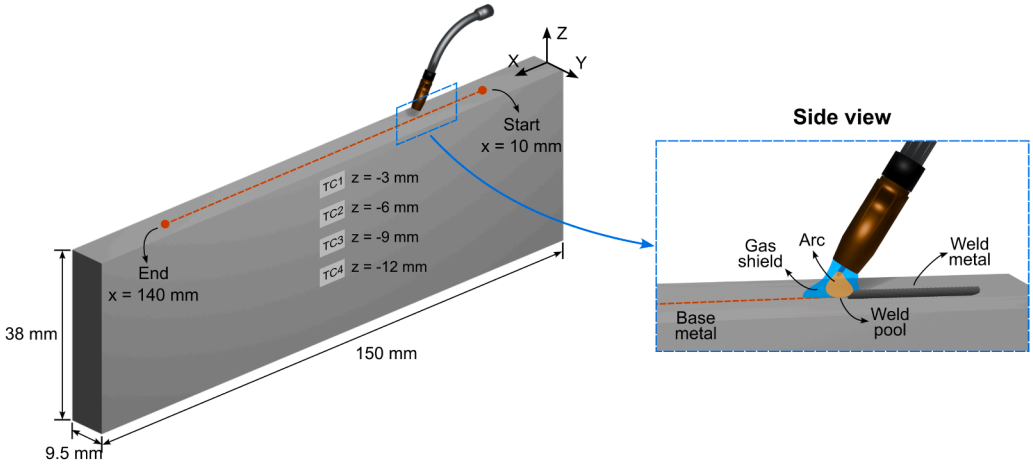


Fig. 2. Sample dimensions, weld line and thermocouple locations. Adapted from [28].

Table 2
Thermal properties of 6061-T6 alloy.

Property	Value
Thermal conductivity (W/mK)	167
Specific heat capacity (J/kgK)	896
Density (kg/m^3)	2700
Melting Point (°C)	650

present study. WelPhys AI is a Python-based framework that employs physics-informed neural networks (PINNs) to solve the transient heat conduction equation associated with a moving heat source, making it particularly suitable for modeling thermal phenomena in MIG/MAG welding. In its current implementation, the framework supports fully three-dimensional, transient simulations, enabling the prediction of the spatial and temporal evolution of temperature fields.

The framework is formulated to support Dirichlet, Neumann, and Robin boundary conditions in a general manner; however, only convective (Robin-type) boundary conditions were applied in the present study to ensure consistency with the experimental setup. The convective heat transfer coefficient and ambient temperature adopted in the simulations were $h = 15.0$ Wm^{-2}K^{-1} and $T_\infty = 295.15$ K, respectively. Initial conditions were uniformly prescribed over the entire computational domain. GPU acceleration was implemented using the PyTorch deep learning library, enabling efficient training through optimization of a composite loss function that enforces the governing physics, boundary conditions, and available data. All simulations were trained on an ASUS TUF Gaming GeForce RTX^{TM} 4090 GPU (10,496 CUDA cores and 24 GB of GDDR6X memory).

This framework builds on the WelPhys Thermal solver (validated for gas metal arc welding-based processes [2,7,29] and wire arc additive manufacturing [30]), which employs the classical finite-volume method to solve the governing heat equation while enforcing thermal energy conservation. After completing the PINN validation against experimental measurements, we additionally benchmark the trained PINN against WelPhys Thermal to quantify computational efficiency. Including time-to-solution, thereby providing a direct comparison with a state-of-the-art, high-fidelity FVM baseline.

5. Results and discussion

5.1. Validation of the PINN

To validate the proposed PINN, the training process was conducted over 60,000 epochs using the Adam optimizer ([31]) with a learning rate of 1×10^{-3}. To enhance convergence, a learning rate scheduler was employed, reducing the learning rate by a factor of 0.5 when the loss plateaued for 100 consecutive epochs. Additionally, gradient clipping was applied with a maximum norm of 1.0 to prevent instability due to exploding gradients. The PINN fully connected architecture consisted of four layers with a structure of [4, 100, 100, 1], where the input layer received four normalized variables (time and spatial coordinates), two hidden layers contained 100 neurons each with hyperbolic tangent (tanh) activation functions, and the output layer predicted the normalized temperature field. All layers were fully connected. Weights and biases were initialized per PyTorch defaults (Kaiming uniform for weights; uniform bias).

The discretization of the computational domain was performed using Latin Hypercube Sampling (LHS) [32], a stratified sampling strategy selected to ensure a space-filling distribution with minimal clustering. The training dataset consists of fixed point counts for each loss component: $N_{ic} = 1,000$ points sampled at $t = 0$ for initial conditions; $N_{bc} = 10,000$ points sampled from the domain surfaces for boundary conditions; and $N_{pde} = 20,000$ interior collocation points for enforcing the governing partial differential equation. To rigorously assess model robustness and mitigate the stochastic influence of neural network weight initialization, the training process was repeated 5 times using distinct random seeds. The model exhibited high stability across these independent runs, achieving a mean l_2 error of 4.21 K with a standard deviation of $\sigma = 0.27$ K. This statistical consistency confirms that the reported accuracy is a fundamental characteristic of the proposed architecture rather than an artifact of a “lucky" initialization.

The LHS method is applied globally across the domain without additional residual-based adaptive refinement or heuristic oversampling near the moving heat source. Furthermore, labeled experimental data from thermocouples at four distinct spatial locations were incorporated into the training process. These measurements, preprocessed through normalization and rescaling, were used to enhance model accuracy by directly minimizing the discrepancy between predicted and observed temperature values.

The training loss history for the PINN model is depicted in Fig. 3. The total loss (blue curve) exhibits a consistent downward trend throughout the 60,000 training epochs, demonstrating effective convergence. Initially, the total loss starts at approximately 10^0 and progressively decreases by more than an order of magnitude, stabilizing near 10^{-2}. The periodic spikes observed in the loss curves correspond to learning rate adjustments triggered by the learning rate scheduler, which ensures continued optimization by reducing the learning rate when the loss plateaus. This strategy effectively prevents stagnation and promotes further refinement of the model’s predictive capabilities.

A breakdown of individual loss components reveals distinct convergence behaviors for each term, which are closely related to the

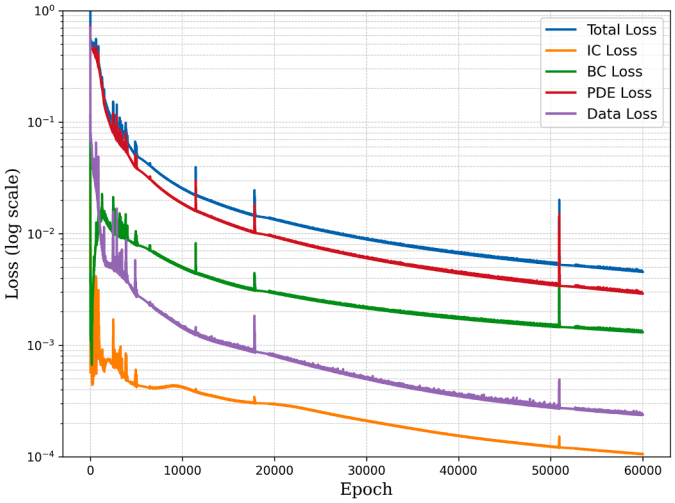


Fig. 3. Training loss of the PINN model.

governing physics of heat transfer in MIG welding. The initial condition loss (orange curve) rapidly declines within the first few thousand epochs, reaching a near-steady state below 10^{-3}. This indicates that the model efficiently captures the initial thermal conditions of the welded plate before the arc initiates. The boundary condition loss (green curve) follows a more gradual decrease, reflecting the model's adaptation to thermal boundary constraints, such as convective heat losses at the metal surface. The physics-based loss associated with the governing partial differential equation (red curve) mirrors the total loss behavior. This loss component is critical in MIG welding simulations, as it ensures that the temperature field correctly responds to the dynamic energy input from the arc and the conduction-dominated cooling process in the metal substrate. This suggests that enforcing the PDE residual is a key factor in driving the network's learning dynamics.

The data loss (purple curve), which quantifies the discrepancy between the predicted and experimental temperature values, also decreases steadily, confirming the positive impact of incorporating thermocouple data into the training process. In the early training stages, the data loss exhibits fluctuations, reflecting the model's adjustment to the non-uniform temperature gradients characteristic of the welding. These gradients arise due to the localized and intense heat input from the arc. Over time, the smooth decline in this loss term highlights the model's ability to assimilate real-world temperature measurements while preserving physical consistency. The final loss values across all components suggest that the trained PINN successfully captures the transient heat dynamics of MIG welding.

Fig. 4 illustrates the PINN predictions compared to experimental data at four thermocouple locations (TC1 - Fig. 4a, TC2 - Fig. 4b, TC3 - Fig. 4c, and TC4 - Fig. 4d), as well as the corresponding absolute error and ℓ_2 norm for each probe. In all cases, the temperature profile increases sharply as the welding arc approaches the thermocouple location and decreases once the arc moves away, reflecting the transient nature of the heat process. The peak temperature values, timing of the temperature rise, and subsequent cooling phases align well with the experimental measurements, reinforcing the model's capability to capture the main physical phenomena of a moving heat source.

The absolute error curves show that the largest deviations occur near the peak-temperature interval. This is consistent with the strongly localized and nonlinear nature of the arc heat input, which produces very steep thermal gradients in the molten pool and adjacent solid material. Elevated discrepancies around the peak are therefore physically expected, since this regime is most sensitive to effects not represented in the present model, such as latent heat of fusion, Marangoni-driven convection within the weld pool, and strongly temperature-dependent thermophysical properties. Even under these rapid transients, the PINN maintains satisfactory accuracy, with errors typically staying below 30 K during the most critical period. Moreover, the thermocouple-wise l_2 norms confirm that the overall mismatch remains low over the full temperature time history.

From a physical point of view, these results demonstrate that the PINN effectively incorporates the PDE describing heat conduction, as illustrated by the 3D temperature fields in Fig. 5. The snapshots in Fig. 5 shows how the maximum temperature region tracks the welding arc's movement over time, forming a pronounced localized hotspot that gradually shifts across the workpiece.

In each of the three time instances shown, the predicted temperature distribution captures both the intense heating near the heat source and the resulting gradients that dissipate into the cooler solid regions. This phenomenon demonstrates the dynamic effects of heat advection and conduction: as the arc moves, fresh material is heated rapidly, while previously heated zones cool and blend with surrounding temperatures.

Furthermore, the contours in Fig. 5 highlight the network's ability to track the peak temperature and capture the steep temperature gradients around the heat source indicates a robust understanding of the welding process mechanics. This synergy between first-principles physics and data-driven learning allows the PINN to not only reproduce steady conduction aspects but also accurately model transient phenomena such as the rapid formation and dissipation of high-temperature regions, ultimately demonstrating its suitability for predictive simulations of MIG welding.

To compare PINN predictions with experimental data, Fig. 6 presents hexbin plots of the PINN-predicted temperatures versus the experimentally measured values at each thermocouple location (TC1 - Fig. 6a, TC2 - Fig. 6b, TC3 - Fig. 6c, and TC4 - Fig. 6d). In all four plots, the diagonal dashed line represents the ideal $y = x$ relationship, indicating good agreement between the predictions and the experimental measurements. A large concentration of points around this diagonal demonstrates that the PINN predictions closely match the measured temperatures across a wide range, from roughly 300 K to about 700 K. The color scale in each plot reflects the local density of data points, highlighting regions where the PINN consistently achieves strong agreement with the experimental data.

From a physics-informed perspective, this close correspondence arises because the network is trained not only on experimental data but also on the governing PDE. The fact that the majority of points cluster along the diagonal line in each subplot implies that the PINN successfully captures these rapid temperature changes. Minor deviations from the diagonal, particularly at higher temperatures, may be attributed to local inhomogeneities in the weld pool, sensor noise, or small variations in welding parameters that are not explicitly modeled.

Incorporating labeled thermocouple measurements into the training process is critical for improving the model's predictive performance. The sensor readings provide direct, high-confidence guidance that steers the PINN's parameters toward the observed temperature trajectories, reducing discrepancies both near the peak-temperature interval and throughout the subsequent cooling phase. In a strictly physics-only setting, the network is constrained only by the PDE residual and boundary conditions, which may not fully account for local or experiment-specific effects, such as small fluctuations in travel speed, subtle spatial variability in material properties, or temporal variations in arc power. By combining sparse experimental measurements with the governing physics, the model learns a more resilient and realistic representation of the welding thermal response.

The strong agreement between predicted and experimental temperature profiles at multiple locations, combined with low ℓ_2 norms and modest maximum errors, confirms that the network accurately captures the spatiotemporal heat transfer behavior induced by a moving arc. Moreover, the model's success at reflecting complex thermal gradients highlights the power of combining first-principles PDEs

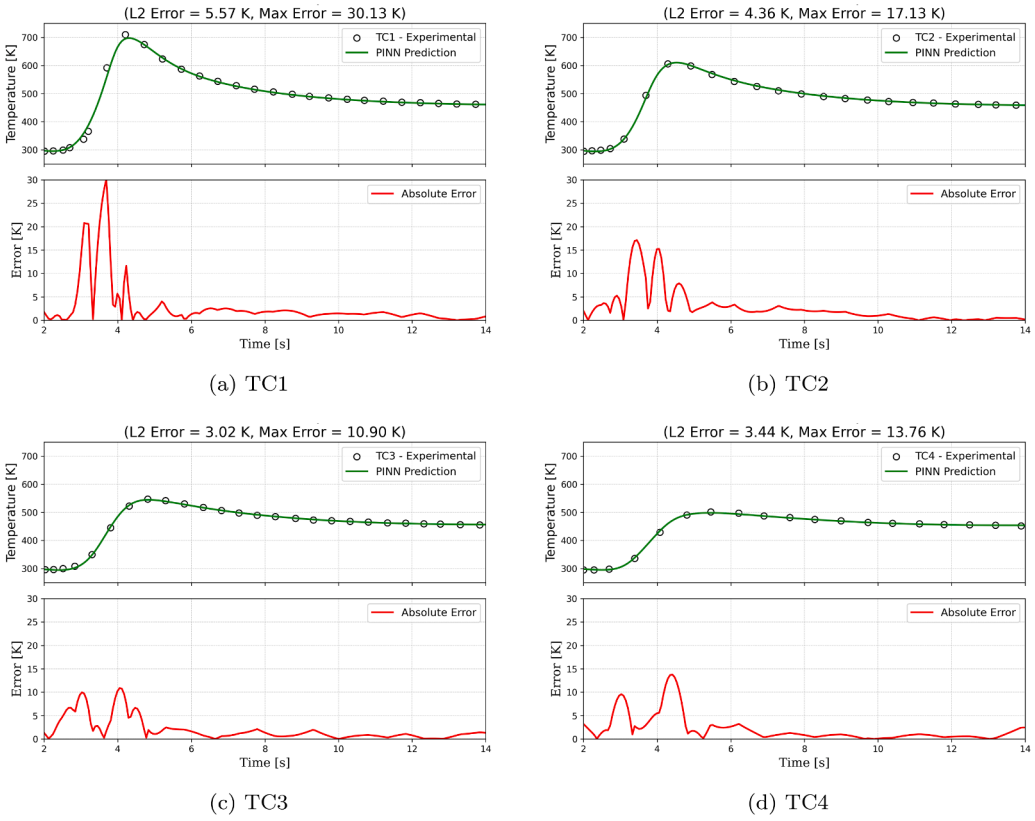


Fig. 4. Comparison of PINN model predictions vs. experimental data [28] at each probe location.

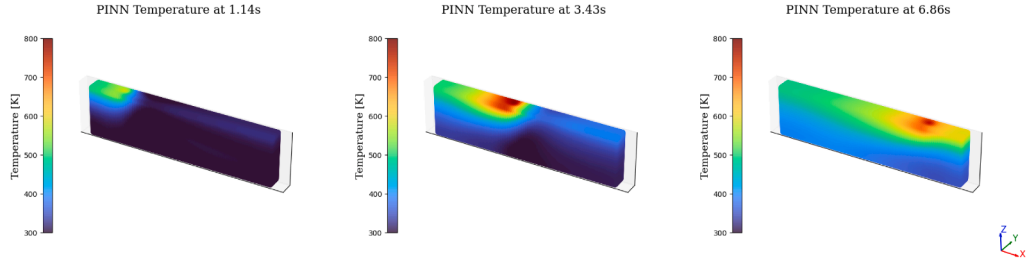


Fig. 5. 3D temperature field of the PINN model at three different time points.

with data-driven corrections in welding simulations. This synergy between physical laws and empirical observations ultimately yields a more reliable and generalizable predictive tool for industrial welding applications.

Table 3 summarizes a reproducible computational benchmark between our finite-volume solver (Welphys Thermal) and the proposed PINN-based surrogate (Welphys AI) for the same welding case and workstation. The FVM simulation required 17 min 18 s to complete a full transient solve on a mesh of approximately 4.5×10^5 control volumes with $\Delta t = 0.02$ s, whereas the PINN converged in 14 min 30 s on an NVIDIA RTX 4090, corresponding to an approximately 16% reduction in the one-time-to-solution for this setting. More importantly, the trained PINN enables millisecond-scale evaluation (< 10 ms per spatio-temporal query), providing direct access to the temperature field at arbitrary points without rerunning a transient simulation. This yields a clear amortized computational advantage in workflows that require many repeated evaluations (i.e., such as parameter studies, uncertainty quantification, optimization loops, or low-latency monitoring/digital-twin deployments) where repeatedly executing a FVM/FEM model becomes impractical.

5.2. Hyperparameter study

In order to assess the influence of hyperparameters on the PINN's ability to predict temperature distributions in MIG welding, a systematic hyperparameter study was conducted. This study is crucial because the performance of PINNs is highly sensitive to hyperparameter choices, which can significantly affect convergence, stability, and the fidelity of temperature predictions. In our investigation, the parameters fixed during the validation process will serve as a baseline, while other key hyperparameters will be varied systematically. Specifically, we will analyze the effect of the learning rate by testing values of 1×10^{-3}, 5×10^{-4}, and 1×10^{-4}; the effect of the epoch size by comparing training runs with 50,000, 100,000, and 200,000 epochs; the effect of network width using architectures of [4, 50, 50, 1], [4, 100, 100, 1], and [4, 200, 200, 1]; and the effect of network depth through architectures of [4, 50, 50, 1], [4, 50, 50, 50, 1], and [4, 50, 50, 50, 50, 1]. By systematically varying these hyperparameters, we aim to identify the optimal configuration that balances computational efficiency with predictive accuracy, thereby enhancing our understanding of how each parameter influences the underlying physics captured by the PINN model.

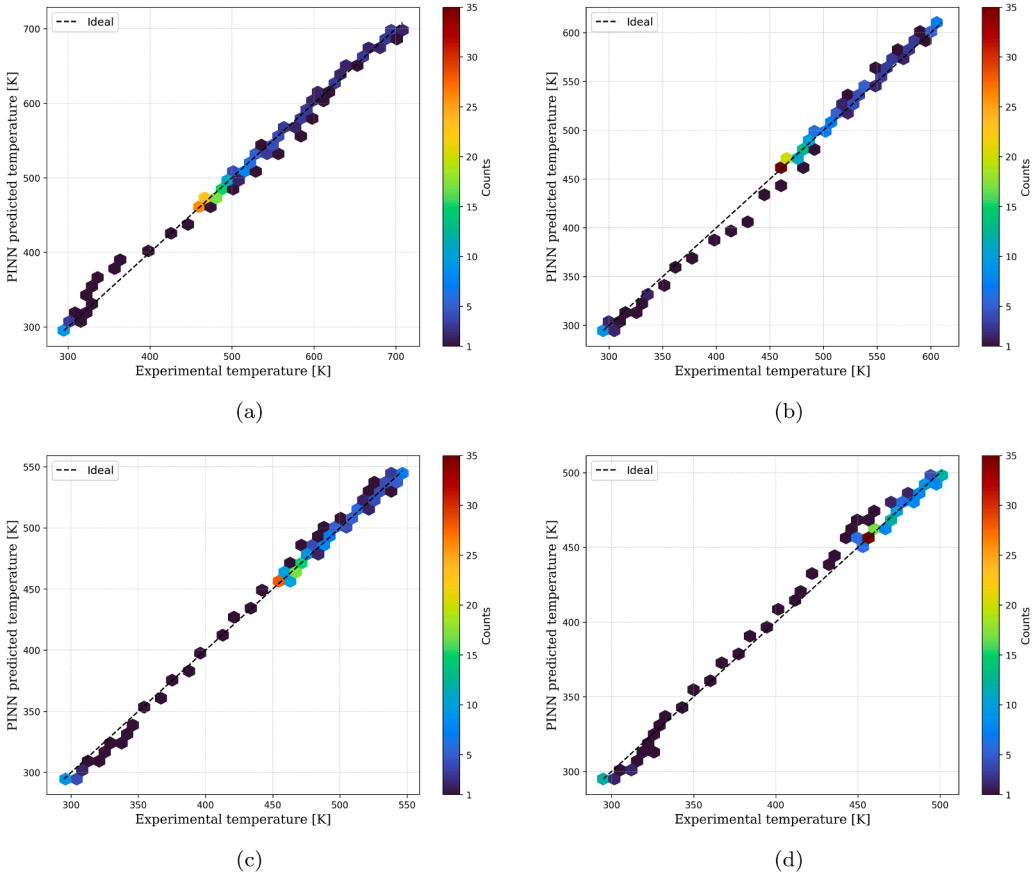


Fig. 6. Hexbin plots comparing PINN-predicted temperatures with experimental measurements at thermocouple locations ((a)-TC1, (b)-TC2(c)-TC3(d)-TC4. The dashed diagonal line indicates the ideal $y = x$ relationship.

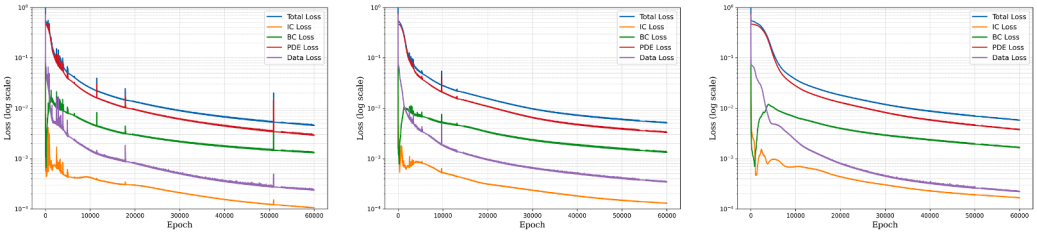
Table 3
Benchmark comparison between the in-house FVM solver (Welphys Thermal) and Welphys AI framework for the same welding case and workstation.

Metric	Welphys Thermal	Welphys AI
Hardware	CPU (Intel Core i9-14900k)	NVIDIA RTX 4090
Model type	Full-order transient solver	Trained surrogate (PINN)
Mesh	$\sim 4.5 \times 10^5$ control volumes	Mesh-free (30k collocation points)
Time step	$\Delta t = 0.02$ s	—
ime to solution	17 min 18 s	14 min 30 s (training)
Inference	Full re-run for new condition	< 10 ms per spatio-temporal inference

5.2.1. *Effect of learning rate*

The learning rate plays a critical role in training stability and convergence speed for the PINN. We compared three learning rates (1×10^{-3}, 5×10^{-4}, 1×10^{-4}) while keeping other settings constant to assess their impact on model accuracy. Generally speaking, all three learning rates eventually allowed the PINN to learn the welding thermal profile, but with noticeable differences in convergence behavior (Fig. 7). The highest learning rate (1×10^{-3}) led to the fastest initial decrease in training loss, rapidly fitting the dominant heating features, whereas the lowest rate (1×10^{-4}) converged more gradually. The inclusion of a learning-rate scheduler and gradient clipping in training was crucial here. For the 1×10^{-3} run (Fig. 7a), these mechanisms curbed instabilities and ensured smooth convergence despite the aggressive updates. In contrast, the smallest learning rate produced very stable but slow improvements in loss, requiring many iterations to reach the same error level. The intermediate rate (5×10^{-4} - Fig. 7b) showed a convergence rate between these extremes, but interestingly did not outperform the highest rate in final accuracy, suggesting it may have gotten stuck in a mild plateau before the scheduler could sufficiently reduce it. This indicates that an overly conservative learning rate can prolong training without guaranteeing better accuracy, while an initially high rate, if properly controlled, can quickly capture the solution's coarse features and then refine them after scheduler-induced reductions.

Despite reaching low loss values in all cases, the choice of learning rate influenced how well the model captured the temperature dynamics of the moving heat source. In MIG welding, each thermocouple sees a sharp temperature rise as the arc approaches and a rapid fall as it moves away. The PINN with a high learning rate adapted quickly to fit this steep temporal gradient, closely matching the timing and height of the heat pulse at the probe. The low learning rate model, by the end of the allotted epochs, had made steady progress but tended to slightly under-predict the peak temperature at the hottest location (TC1 - Table 4), likely because it had not fully converged within the same training budget. Notably, the moderate learning rate (5×10^{-4}) gave the largest error at TC1. The l_2 error at this critical sensor was 7.16 K for 5×10^{-4}, compared to 5.57 K for 1×10^{-3}. This intermediate rate evidently did not optimize the network parameters as effectively for the high-gradient region, possibly because it was neither fast enough to



(a) 1×10^{-3} (b) 5×10^{-4} (c) 1×10^{-4}

Fig. 7. Training loss of the PINN model for each learning rate.

Table 4
Experimental [28] vs. PINN peak temperatures and error metrics at each learning rate (in K).

Thermocouple	Learning rate	Experimental	PINN prediction	l_2 Error	Max. absolute error
TC1	1×10^{-3}	709.7	697.8	5.572	30.131
	5×10^{-4}	709.7	694.2	7.157	37.210
	1×10^{-4}	709.7	698.0	5.970	34.033
TC2	1×10^{-3}	605.6	610.4	4.357	17.126
	5×10^{-4}	605.6	609.7	4.459	18.970
	1×10^{-4}	605.6	611.0	4.667	19.509
TC3	1×10^{-3}	546.8	544.7	3.024	10.896
	5×10^{-4}	546.8	544.8	2.877	11.442
	1×10^{-4}	546.8	543.7	3.105	13.874
TC4	1×10^{-3}	501.4	498.2	3.442	13.765
	5×10^{-4}	501.4	498.8	3.633	12.225
	1×10^{-4}	501.4	497.6	3.411	14.973

quickly capture the peak during early training, nor slow enough to fine-tune with many small updates, a dynamic that indicates the nonlinear effect of learning rate on convergence. By contrast, the highest learning rate model achieved the lowest error at TC1 (and also at TC2) among the three, indicating that it managed to learn the heat accumulation at the weld arc more accurately within the given epochs. All models, however, maintained reasonably low errors across the cooler thermocouples (TC3, TC4), with differences on the order of only a few tenths of a Kelvin in l_2 error, suggesting that learning rate choice had a more pronounced impact on fitting the extreme thermal conditions than on the milder, diffusion-dominated regions.

The differences in predictive performance become evident when examining the temperature profile at TC1. Fig. 8 (based on TC1 data) shows that all three learning rates yield PINN predictions that follow the general experimental curve, a sharp rise to a peak of around 700 K followed by a cooling tail, but the alignment with the measured peak and the subsequent cooling trajectory varies. The 1×10^{-3} PINN tracks the thermocouple data most closely, peaking at nearly the correct temperature and time, whereas the 5×10^{-4} case exhibits a slight lag and underestimation of the peak (its maximum absolute error at TC1 is 37 K, worse than the 30 K error with 1×10^{-3}). The lowest learning rate 1×10^{-4} also under-shoots the peak somewhat (max error 34 K at TC1), although its overall error is lower than the 5×10^{-4} case, implying that given more training iterations it would likely catch up. Importantly, all three runs capture the qualitative physics: a steep temperature gradient when the heat source is nearby and a gradual decay as heat dissipates into the workpiece. This is a testament to the PINN's physics-informed foundation, even when training is not fully optimal, the governing PDE in the loss function constrains the solution to behave physically. The high learning rate model, benefiting from swift initial learning, conforms very well to the expected thermal response, while the low learning rate model demonstrates the same trend but with a mildly damped peak due to incomplete convergence. Meanwhile, the 5×10^{-4} model's underperformance at the peak suggests that it may require either more epochs or a more aggressive reduction schedule to fine-tune the high-temperature region.

Despite these small differences, it is worth noting that the PDE-driven training provided a strong regularization: none of the models exhibited

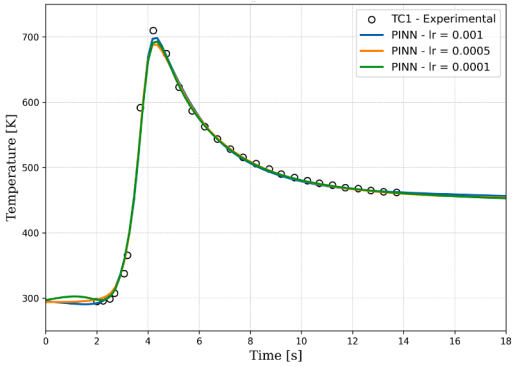
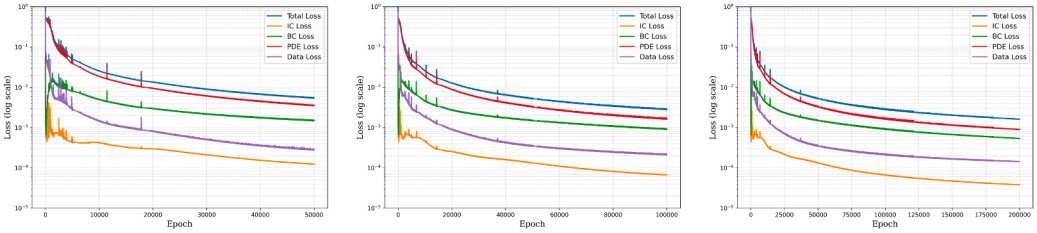


Fig. 8. Comparison of PINN model predictions at varying learning rates with experimental data [28] at TC1 probe location.

non-physical behavior (such as oscillatory or divergent temperature predictions) even with the highest learning rate, thanks to the heat equation acting as a guide rail that keeps the solution physically plausible. The learning rate primarily affected how tightly the PINN fit the experimental data versus the PDE in the limited training time. Taken together, the study indicates that a relatively higher initial learning rate (on the order of 10^{-3}) with proper scheduling is advantageous for this problem, as it swiftly captures the peak heat accumulation and associated steep gradients from the moving welding arc, whereas an overly low or moderate learning rate may leave some of these transient features under-fit by the end of training. The training time measurements reinforce this conclusion (Table 5): the 1×10^{-3} run actually finished slightly faster than the others (due to earlier achievement of loss plateaus and consequent learning rate reductions), highlighting that no significant penalty was paid in computational cost. Finally, tuning the learning rate is vital, it must be high enough to track the fast thermal dynamics of MIG welding, yet controlled enough to maintain stability.

5.2.2. *Effect of epoch size*

The total number of training epochs was varied (50k, 100k, 200k) to evaluate how prolonged training influences convergence stability and



(a) 50000 epochs (b) 100000 epochs (c) 200000 epochs

Fig. 9. Training loss of the PINN model for each epoch.

Table 5
Training times for each different PINN learning rate configuration.

Learning rate	Training time (s)
0.001	846.27
0.0005	880.28
0.0001	881.11

the model's ability to capture the full temporal heat cycle. As expected, more training epochs yield a lower final loss and improved fit to both the physics and data (Fig. 9). Extending the training from 50k to 200k epochs produced a monotonic decrease in the total loss, indicating that the PINN continued to refine its solution and better satisfy the governing heat equation and measurement constraints with additional iterations. However, the benefits of extra epochs showed diminishing returns beyond a certain point: the loss reduction curve flattened noticeably after 100k epochs. In practical terms, the model trained to 100k epochs was already very close to convergence, and pushing to 200k epochs resulted in only marginal further loss improvement. This behavior suggests that by 100k epochs the network had learned most of the underlying solution structure, and further training primarily fine-tuned smaller details. The convergence trends were consistent and stable, no divergence or instability was observed even at 200k epochs, emphasizing that the PINN optimizer can steadily approach a physical solution given sufficient time. The enforcement of the PDE in the loss likely contributed to this stability, as it prevents the network from straying into unphysical function space regions and thus makes the long training process focus on genuine error reduction rather than oscillatory adjustments.

Compared with the experimental data, most thermocouple locations saw error reductions when going from 50k to 100k to 200k epochs (Table 6). For example, at TC4, the l_2 error dropped from 3.98 K after 50k epochs to 3.05 K at 100k, and further to 2.28 K at 200k. This indicates a progressively better capture of the thermal propagation into the far reaches of the plate with more training, the PINN fine-tuned how the heat diffuses to distant points over time. Similarly, the largest temperature peak (at TC1) was captured more accurately with extended training: the maximum absolute error at TC1 decreased dramatically from about 30.8 K with 50k epochs to 18.3 K with 200k epochs. The l_2 error at TC1 improved from 5.47 K to 3.92 K over the same range. This means the model's representation of the heat at the weld center became much closer to reality as training progressed. Interestingly, one sensor (TC2) did not follow the monotonic improvement trend: the l_2 error at TC2 was 3.91 K at 50k, which slightly increased to 4.16 K at 100k and 4.22 K at 200k. This minor fluctuation (within about 0.3 K) is likely within the noise or uncertainty range, and might indicate that after extensive training the model redistributed error among locations (possibly focusing on reducing TC1 and TC4 error a bit at the expense of a tiny increase at TC2). Overall, the global accuracy trend was positive with more epochs, and no evidence of overfitting to the thermocouple data was observed even at 200k epochs. Thanks, in part, to the regularizing

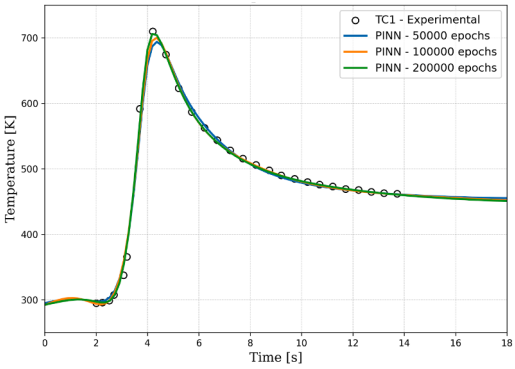


Fig. 10. Comparison of PINN model predictions at varying epoch sizes with experimental data [28] at TC1 probe location.

effect of the physical loss, the PINN did not start fitting spurious features but rather kept improving the physical fidelity of the solution.

A key aspect influenced by training duration is the temporal capture of the heat at each sensor (Fig. 10). With only 50k epochs, the PINN had learned the general shape of the temperature vs. time curve but small discrepancies were evident. For instance, the peak temperature at TC1 was under-predicted (694.8 K vs 709.7 K actual) and the timing of the peak and cooldown may have been slightly offset. By increasing to 100k epochs, these discrepancies diminished: the TC1 peak prediction rose to 700.0 K, much closer to the true value, and the overall curve alignment improved. At 200k epochs, the TC1 peak reached 704.2 K, just 5 K below of the experimental peak. The model effectively converged to the correct amplitude of the heat accumulation event with sufficient training. With limited training, the network may not perfectly balance the conduction rates in the model, but extended training ensures that the PDE constraint is satisfied more tightly, leading to the correct diffusion behavior.

While longer training clearly yields a more accurate model, it comes with significant computational cost (Table 7). The runtime scales roughly linearly with epoch count. Our simulations showed training times of 683 s for 50k epochs, 1724 s for 100k, and 2841 s for 200k on the given hardware. Thus, doubling the epochs from 50k to 100k nearly tripled the training time, and doubling again to 200k added another 65% on top of that. From a practical standpoint, 100k epochs emerged as an excellent compromise between accuracy and efficiency. The model at 100k epochs had already achieved a large portion of the accuracy gains (e.g. cutting TC1 peak error almost in half compared to 50k) while using 40% less computation time than the 200k-epoch model. In many engineering applications, such a trade-off is desirable: the slight accuracy sacrifice (on the order of a few Kelvin error in worst-case) might be acceptable if it saves a significant amount of training time. Moreover, the stability of the PINN solution at 100k was very good. The loss had leveled off, indicating the solution was near stationarity with respect to the optimization. Additional epochs mostly fine-tuned the solution to reduce the residual error. We also note that beyond a certain point,

Table 6
Experimental [28] vs. PINN peak temperatures and error metrics by epoch (in K).

Thermocouple	Epochs	Experimental	PINN prediction	l_2 Error	Max. absolute error
TC1	50,000	709.7	694.8	5.473	30.827
	100,000	709.7	700.0	4.387	24.272
	200,000	709.7	704.2	3.920	18.282
TC2	50,000	605.6	609.7	3.910	16.852
	100,000	605.6	611.3	4.161	19.525
	200,000	605.6	611.1	4.223	19.948
TC3	50,000	546.8	544.5	2.991	10.927
	100,000	546.8	544.3	2.772	9.567
	200,000	546.8	543.9	2.311	9.696
TC4	50,000	501.4	498.0	3.981	14.398
	100,000	501.4	497.5	3.049	11.503
	200,000	501.4	499.6	2.284	9.692

Table 7
Training times for each different PINN epoch configuration.

Epochs	Training time (s)
50,000	683.18
100,000	1724.17
200,000	2841.06

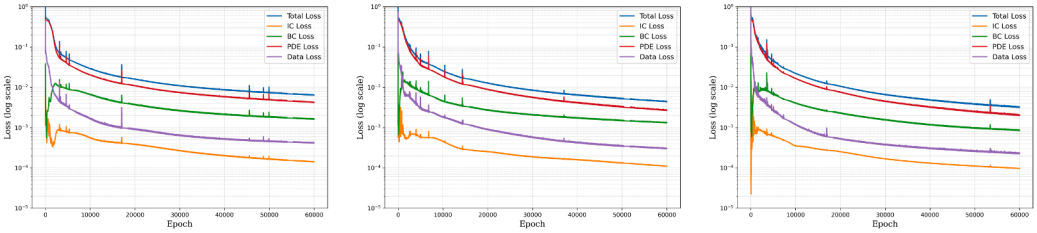
continuing training risks encountering the limits of measurement noise or unmodeled physics: for example, the slight rise in TC2 error could be a sign that the model started to emphasize fitting the TC1 and TC4 data and the PDE residual so much that it slightly deviated at TC2, which might already be within experimental uncertainty. Thus, exceedingly long training might lead to diminishing or even negative returns if the model starts to chase noise. In our case 200k did not overfit in a classic sense, the model remained physically reasonable and accurate, but the improvement from 100k to 200k was minor. Therefore, for generalization and efficiency, one might stop training around 100k once the loss curve flattens, relying on the physics prior to have guided the solution to a good optimum by that time.

5.2.3. *Effect of network width*

The network width strongly affects the PINN's capacity to approximate the complex temperature field in welding. In this sense, we investigated three architectures with increasing width: [4, 50, 50, 1], [4, 100, 100, 1], and [4, 200, 200, 1] (all with two hidden layers of the specified neuron count - Fig. 11). Intuitively, a wider network can represent more complex functions, which is important for capturing the sharp thermal gradients and localized heating in this problem. The MIG welding temperature field has steep, high-temperature regions near the moving arc and more gradual variations further away; an under-parameterized network may smooth out or miss some of these features. Our results indeed show that increasing the width significantly improves the model's accuracy up to a point. Expanding each layer from 50 to 100 neurons produced a dramatic drop in error across all thermocouple signals, indicating that the 50-neuron network lacked sufficient complexity for this task (Table 8). For instance, the l_2 error at TC1 (the sensor nearest the weld path) was reduced from 10.69 K with 50 neurons to 5.48 K with 100 neurons, nearly a 50% reduction. Similar large improvements were seen at other probes (TC2-TC4) when going to 100 neurons, confirming that the 50-wide network was under-fitting the temperature distribution. This under-fitting is especially pronounced in regions with steep thermal gradients: the 50-neuron model could not fully capture the extremely localized heat input of the arc, resulting in higher errors particularly around the weld area and moments of peak heating. By contrast, the 100-neuron model had enough representational power to approximate the rapid temperature rise and fall more faithfully, as well as the spatial temperature gradients emanating from the weld line.

Pushing the width further from 100 to 200 neurons per layer yielded surprisingly minimal gains. The total loss did decrease slightly and the TC1 error marginally improved (from 5.48 K to 5.36 K in l_2), but this difference is very small. In some cases, the 200-neuron network even performed a bit worse at certain locations: for example, the l_2 error at TC2 increased from 4.23 K with 100 neurons to about 4.58 K with 200 neurons. These differences are minor (on the order of a few tenths of a Kelvin), but they suggest that simply adding more neurons beyond a moderate width doesn't necessarily improve the PINN's ability to model the welding physics. It may even introduce slight overfitting or unnecessary complexity that manifests as local performance fluctuations. Essentially, once the network was wide enough to capture the primary features of the solution (100 neurons was sufficient here), extra width gave diminishing returns. The 100-neuron architecture already had the capacity to represent the steep temperature gradients and the smooth diffusion tails, so doubling the neuron count only marginally improved the fit at the hottest spot (TC1) and did not uniformly improve all areas of the solution (Fig. 12). This is a common scenario in function approximation: beyond a certain model complexity, the error is dominated by factors like data noise or optimization limits, not the expressive power of the network. In our PINN, the physics prior further discourages overly complex or non-physical functions, so an excessively wide network might be slightly harder to train optimally (more parameters to adjust under the same loss constraints) and could focus on tiny residual errors that are not crucial, hence the slight degradation at TC2 when using 200 neurons.

The influence of width is particularly evident in how well the model assimilated the TC1 thermocouple data, which reflect the highest temperature and most rapid changes. A narrower network (50 neurons) struggled to fit the peak temperature at TC1, predicting a maximum temperature of only 692.5 K at that location. This is about 17 K lower than the experimental peak of 709.7 K, a clear underestimation of the heat accumulation. With 100 neurons, the PINN's predicted peak at TC1 rose to 694.0 K, and with 200 neurons it increased further to 698.2 K. Thus, increasing width closed the gap between predicted and measured peak temperature, an indication that the heat accumulation near the weld arc is better captured by a higher-capacity model. The maximum error at TC1 correspondingly dropped from a high 39 K (50 neurons) to 30.7 K (100 neurons) and 26.5 K (200 neurons). In practical terms, the 50-neuron network likely produced a smoother, flatter temperature profile at TC1 (unable to reach the true peak), whereas the 100-neuron network could form a sharper temperature spike, much closer to the actual thermocouple reading. The 200-neuron network sharpened this a bit more, but by that point the model was only about 1.5 K better in l_2 error at TC1 than the 100-neuron case. This confirms that the majority of the improvement came from correcting the gross underestimation that the narrow network had. The ability to fit the steep temperature gradient at the moment the arc passes (which causes the rapid 400 K rise at TC1) was significantly enhanced by going to 100 neurons. Notably, even the



(a) 50 neurons (b) 100 neurons (c) 200 neurons

Fig. 11. Training loss of the PINN model for each neuron size.

Table 8
Experimental [28] vs. PINN peak temperatures and error metrics by neuron count (in K).

Thermocouple	Neurons	Experimental	PINN prediction	l_2 Error	Max. absolute error
TC1	50	709.7	692.5	10.691	39.001
	100	709.7	694.0	5.484	30.728
	200	709.7	698.2	5.362	26.482
TC2	50	605.6	611.1	7.730	21.157
	100	605.6	611.0	4.232	18.995
	200	605.6	612.4	4.580	19.573
TC3	50	546.8	546.6	5.541	16.082
	100	546.8	545.7	3.282	10.503
	200	546.8	545.2	2.738	11.712
TC4	50	501.4	497.7	4.951	16.249
	100	501.4	497.2	3.884	13.537
	200	501.4	498.0	2.750	13.155

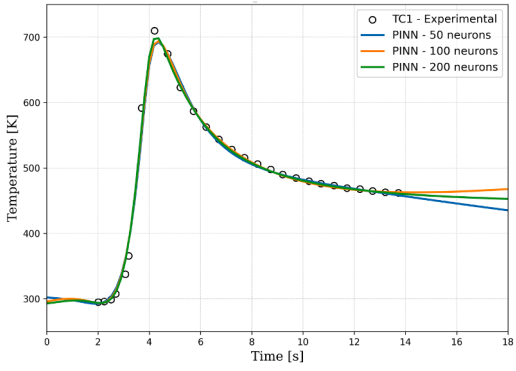


Fig. 12. Comparison of PINN model predictions at varying neuron sizes with experimental data [28] at TC1 probe location.

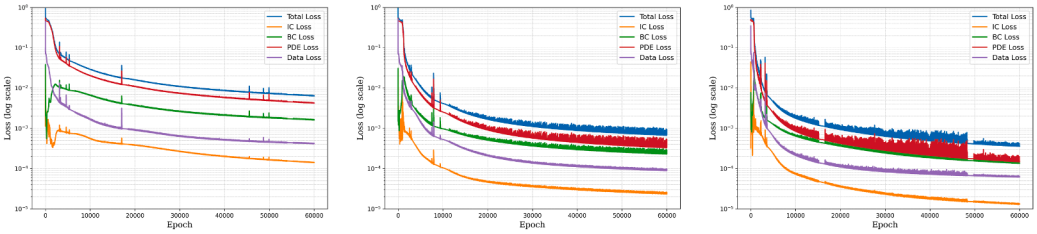
Table 9
Training times for each different PINN neuron configuration.

Neurons	Training time (s)
50	829.73
100	1013.25
200	1366.26

far-field thermocouple TC4 saw its l_2 error drop from 4.95 K to 3.88 K when increasing width from 50 to 100, showing that the wider network also improved the modeling of cooler, diffusion-dominated regions. This suggests that the additional neurons helped in all aspects: the network could allocate some capacity to accurately capturing the intense local heating and some to capturing the broader conductive heat spread.

From a physics standpoint, capturing thermal diffusion gradients and sharp spatial temperature changes requires sufficient function complexity. The weld process creates a very localized high-temperature zone and steep spatial gradients near the arc. The 50-neuron PINN was too simplistic to represent the full steepness of these gradients, resulting in larger errors especially during the peak heating period. By contrast, the 100-neuron PINN managed to learn the steep temperature distribution around the weld pool. For example, it could more precisely capture how the temperature drops off within a few millimeters from the arc. This is reflected in its lower errors and the fact that the predicted hotspot traveled and decayed in a way much closer to the experimental data. The wider networks also better honored the physics of heat conduction in that they could model a very high gradient near the heat source and simultaneously maintain smooth, physically plausible transitions to the cooler regions.

Considering both accuracy and efficiency, 100 neurons per layer is the optimal width for this PINN model under our conditions. The 50-neuron network was clearly under-capacity, and the 200-neuron network yielded negligible accuracy gains for a substantial increase in computational cost (Table 9). Training times rose from 830 s at 50 neurons to 1013 s at 100 neurons, and further to 1366 s at 200 neurons. The jump from 100 to 200 neurons increased training time by 35% for only a very slight error reduction. In contrast, going from 50 to 100 neurons increased training time by 22% but cut errors almost in half, a very worthwhile trade-off. The results highlight an important principle: one should choose a network width that is just sufficient to model the complexity of the physics at hand. In our case, once the network was wide enough to represent the multi-modal temperature field (which includes both sharp peaks and smooth backgrounds), making it wider yielded no significant new insights. The PDE enforcement in the PINN inherently guides the network towards physically consistent solutions, so a moderate-size network can already excel if it aligns well with the underlying physics. Excessive width, on the other hand, may lead to redundancy and slight overfitting of minor details. Therefore, for this welding application, [4, 100, 100, 1] can be considered an efficient and effective architecture, balancing the need to resolve steep thermal gradients and trailing diffusion tails against the desire for shorter training times.



(a) 2 hidden layers (b) 3 hidden layers (c) 4 hidden layers

Fig. 13. Training loss of the PINN model for each hidden layer.

5.2.4. Effect of network depth

Finally, we examined the effect of network depth by varying the number of hidden layers in the PINN architecture. Three configurations were tested: two hidden layers ([4, 50, 50, 1]), three hidden layers ([4, 50, 50, 50, 1]), and four hidden layers ([4, 50, 50, 50, 50, 1]), keeping the neuron count per layer at 50 for a fair comparison. Deeper networks can, in principle, learn more hierarchical features of the solution, which is pertinent for modeling multiscale thermal phenomena. In the context of MIG welding, the temperature field indeed exhibits multi-scale characteristics: there is a highly localized, rapid heating at the weld arc superimposed on a broader, slower evolving heat distribution as heat conducts through the workpiece. A shallow network might have difficulty capturing these disparate scales simultaneously, whereas a deeper network might dedicate different layers to different scales or complexities of the function (for example, some layers could learn the coarse background temperature rise, and others the fine details of the peak and its immediate decay). Our findings support this intuition, increasing the depth greatly improved the PINN's accuracy up to three hidden layers, after which gains plateaued (Fig. 13).

There was a significant jump in accuracy from 2 layers to 3 layers, demonstrating the value of additional depth (Table 10). The 2-hidden-layer network (which, recall, had 50 neurons per layer) had the highest errors, comparable to the 50-wide network discussed earlier. For instance, it had an l_2 error of 10.69 K at TC1, indicating considerable mismatch at the peak. When a third hidden layer was added (3 layers of 50), the PINN's errors dropped sharply across all thermocouples. The TC1 l_2 error fell to 5.85 K (from 10.69 K), nearly identical to the error achieved by the much wider 100-neuron network, showing that increasing depth can be as effective as increasing width for capturing complex behavior. All sensors saw improved predictions: the 3-layer model clearly learned the welding heat distribution much better than the 2-layer model. This suggests that the 2-layer network was not expressive enough in depth to capture the temporal or spatial complexity of the solution - likely it struggled with the simultaneous representation of the sharp near-field temperature spike and the smooth far-field conduction profile. The third layer gave the network additional degrees of freedom to shape the function, yielding a solution that fits both the steep gradients and the diffuse regions more accurately. In terms of training dynamics, the deeper networks achieved lower final losses than the shallow one, which implies they found a better optimum in satisfying the PDE and data. This aligns with theory: extra layers allow the network to untangle complex mappings more effectively and can help navigate the optimization landscape by breaking the problem into more intermediate transformations.

Going from 3 hidden layers to 4 hidden layers yielded smaller, incremental benefits. The 4-layer network did perform slightly better on some metrics. For example, it further reduced the l_2 error at TC1 from 5.85 K to 5.6 K, and notably improved predictions for thermocouples farther from the weld. The result highlights that adding the fourth layer provided further improvements especially for TC3 and TC4 (the far-field probes) which experience more diffuse thermal fields. In other words, the deepest network was marginally better at capturing the broad, low-gradient temperature distribution away from the arc. This makes sense: the far-field temperature rises only a little (i.e., hundreds of degrees lower than the weld pool) and is influenced by cumulative conduction over time. A deeper network might capture such long-range diffusion effects more accurately, perhaps by dedicating part of the network to modeling the slowly varying background temperature increase. The 3-layer network was already quite good, but the 4-layer network refined those diffuse predictions further, bringing, say, TC4's error down by a small amount. However, the improvements in the near-field (e.g., TC1, TC2) were very slight with the fourth layer, indicating that three layers were already sufficient to model the highly localized phenomena. This diminishing return with depth mirrors what we saw with extreme width. Once the network has enough complexity to handle the key features, extra complexity yields only minor gains. It's also worth noting that extremely deep networks can be harder to train due to vanishing gradients or getting stuck in local minima, although in our case the difference between 3 and 4 layers in training convergence was not dramatic (both converged to low losses, Fig. 13). We did observe an anomaly in recorded training times: the 4-layer network trained in slightly less time than the 3-layer, which is likely due to run-to-run variance rather than an inherent advantage. Typically, more layers would mean more parameters and thus longer training for the same number of epochs. In any case, the optimization landscape did not seem to pose insurmountable issues for the deeper networks. They successfully converged to lower losses than the shallow network, confirming that depth was a help, not a hindrance, in this physics-informed learning scenario.

The benefits of increased depth can be interpreted through the lens of multiscale thermal phenomena. MIG welding involves a moving heat source (providing a sort of advective heat transport) and heat dissipation through conduction (a diffusive process). These two aspects occur at different scales: the arc introduces very rapid, localized temperature spikes, while diffusion causes a slower, broader spread of heat. A deeper PINN appears to handle this composite behavior more adeptly. We can surmise that the network's first few layers learn to approximate smoother components of the solution (e.g., the overall rise of the plate's temperature field as it is heated over time), while subsequent layers capture finer details (e.g., the sharp temperature rise right under the arc and the steep gradient at the edge of the weld pool). In effect, additional layers provide a form of functional refinement or hierarchical feature learning, where the PINN builds the solution progressively, a coarse approximation in early layers, with later layers adding corrections that account for high-frequency or highly localized features. The result is a closer match to the true temperature field at all scales. The evidence of this is seen in the fact that the 4-layer network best captured the diffuse thermal tails, without compromising the accuracy of the near-field peak. Likewise, the 3-layer network was a big step up in capturing the intense, rapid thermal cycles at the weld (reducing peak errors drastically), which the 2-layer network had trouble with. Essentially, depth gave the PINN the flexibility to simultaneously satisfy the heat equation globally and the local data trends. It could fit the rapid temperature changes induced by the moving arc and the smooth cooling trends governed by conduction. This is particularly important for trailing thermal gradients behind the heat source: as the weld arc moves on, it leaves behind a hot region that

Table 10
Combined results: experimental vs. PINN peak temperatures and error metrics by number of hidden layers (in K).

Thermocouple	Hidden layers	Experimental	PINN prediction	l_2 Error	Max. absolute error
TC1	2	709.7	692.5	10.691	39.001
	3	709.7	710.2	5.854	27.500
	4	709.7	714.3	5.676	35.090
TC2	2	605.6	611.1	7.730	21.157
	3	605.6	612.4	5.352	21.699
	4	605.6	612.3	5.009	22.641
TC3	2	546.8	546.6	5.541	16.082
	3	546.8	544.6	2.851	8.688
	4	546.8	545.4	2.317	9.577
TC4	2	501.4	497.7	4.951	16.249
	3	501.4	499.2	2.475	10.938
	4	501.4	500.0	2.097	9.464

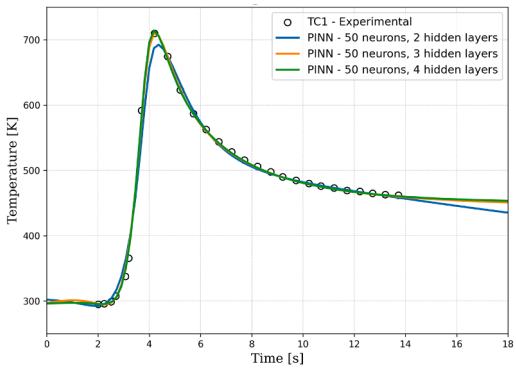


Fig. 14. Comparison of PINN model predictions at varying hidden layers with experimental data [28] at TC1 probe location.

Table 11
Training times for each different PINN hidden layer configuration.

Hidden layers	Training time (s)
2	786.35
3	1248.62
4	1179.75

cools and merges with the surrounding cooler material. A deeper network can represent this process (fast drop near the weld, slow diffusion into the far zone) with greater fidelity. In contrast, a shallow network tends to represent only an average behavior and might blur the distinction between the immediate post-arc cooling and the far-field baseline, leading to larger errors either at the peak or in the tail (Fig. 14). The deeper networks, by virtue of their additional layers, better captured these multi-temporal dynamics. A conclusion supported by their lower errors and the observation that deeper models attained lower PDE residuals (indicating they satisfied both fast and slow dynamics of the PDE more closely).

When weighing the accuracy gains against complexity, the 3-layer network emerges as the most prudent choice. It provided a major improvement over the 2-layer model for a moderate increase in training time (from 786 s to 1249 s, roughly 60% more - Table 11). The 4-layer model was only marginally more accurate while being more complex and computationally heavier (training time 1180 s, on par with 3-layer). The slight improvement with the fourth layer may not justify the added complexity for most applications, as noted in our results. Therefore, unless one needs the absolute lowest possible error and is willing to accept diminishing returns, a 3-hidden-layer PINN (with a suitable width like 100 neurons) is recommended for capturing the welding thermal behavior. This architecture strikes a good balance: it is deep enough to model the advective heat transport and diffusive heat spreading effectively, but shallow enough to train efficiently and avoid overfitting. It's remarkable that the 2-layer network, despite its higher errors, still learned the basic phenomena, which again is due to the physics-based loss keeping it in check. Yet, quantitatively, it could not match the measurements closely. The lack of depth meant it had to compromise, perhaps smoothing out peaks or misaligning timing slightly. By adding a third layer, the PINN no longer had to compromise; it could meet the multiple demands of the problem (initial conditions, boundary conditions, PDE, and data at four locations) much more adeptly. The outcome is a model that generalizes well across the entire domain: it not only fits the training thermocouple data but also respects the physics everywhere, yielding accurate temperature predictions even at points and times where no data were directly given. This reflects the guiding role of PDE enforcement in generalization, with adequate network depth and width, the PINN finds a solution that lies at the intersection of data and physics, ensuring realistic behavior even in between data points. In conclusion, depth is a powerful hyperparameter for PINNs in welding simulations: increasing it to a moderate level (3 layers) greatly enhances the model's ability to capture the multiscale thermal phenomena, while excessive depth returns less benefit. The chosen 3-layer, 100-neuron architecture with a well-tuned learning rate and sufficient epochs was ultimately able to reliably reproduce the complex 3D transient temperature field of the MIG welding process, validating the efficacy of the hyperparameter choices in guiding the PINN toward the true physical solution.

6. Discussion and guidelines for use PINN to predict thermal behavior in MIG/MAG welding

The successful training of the PINN, evidenced by the low final loss values and accurate temperature predictions, demonstrates the network's ability to learn a continuous spatiotemporal solution of the governing transient heat-conduction equation with a moving heat source. Importantly, this physics-constrained formulation yields a differentiable surrogate that preserves physical consistency while avoiding traditional time-stepping and mesh-based spatial discretization.

Beyond reproducing the temperature field, the scientific and engineering value of the proposed framework lies in what this differentiable, mesh-free surrogate enables relative to existing high-fidelity solvers. First, the validated model required 14 min and 30 s of training on a single high-end GPU, after which evaluations at new spatiotemporal points are effectively instantaneous (millisecond-scale per inference). In contrast, the 3D transient FVM simulations required 17 min and 18 s, and coupled thermo-mechanical models can take substantially longer. This shift, from 17 min and 18 s per run to milliseconds per inference after a one-time training cost, changes the feasible workflow: instead

of running a small number of expensive simulations, practitioners can perform large numbers of model evaluations, enabling design-space exploration, uncertainty quantification, and iterative optimization under practical time constraints. In addition, because the PINN output is differentiable with respect to inputs (space, time, and process parameters when included), it can be integrated into gradient-based optimization or control pipelines, which is generally nontrivial with conventional FEM/FVM simulations.

From an engineering point of view, this capability directly supports Industry 4.0 use cases where latency and throughput are critical. For example, the millisecond inference time makes it feasible to deploy the trained model for real-time prediction of peak temperature during welding, using streaming sensor inputs (e.g., travel speed, power, and torch position) to update thermal predictions on-the-fly. Such low-latency prediction is a prerequisite for closed-loop control, where parameters are adjusted during the process to maintain a target thermal cycle, an approach that is computationally impractical if each update requires launching a full 3D transient FEM/FVM solve. Moreover, by enabling rapid virtual trials, the framework can reduce the number of physical experiments required during procedure qualification and parameter tuning, lowering material, labor, and machine-time costs while accelerating development cycles.

Importantly, the developed framework has several direct applications for manufacturing professionals:

- **Real-time process optimization and closed-loop control:** Because inference is fast enough to run at or near process time scales, the trained PINN can be integrated into a supervisory controller to predict thermal metrics (e.g., peak temperature, cooling rate, or a proxy for HAZ width) and adjust welding power or travel speed in real time to track a desired thermal cycle. This enables adaptive welding strategies that are not practical with conventional FEM/FVM due to latency.
- **Rapid virtual trials for cost reduction:** A welding engineer can use the surrogate to simulate large batches of "what-if" scenarios (e.g., varying travel speed, power input, or heat-source parameters) to screen parameter ranges before committing to experiments. This reduces the number of physical trials needed to achieve acceptable HAZ size and thermal histories, lowering consumable usage and minimizing machine downtime.
- **Surrogate-assisted parameter optimization:** The combination of fast evaluation and differentiability makes the model suitable for embedding in optimization loops (e.g., gradient-based methods or evolutionary algorithms) to automatically identify parameters that minimize peak temperatures, control cooling rates, or meet constraints tied to microstructure and residual stress outcomes. Compared with FEM-driven optimization, the proposed approach can reduce the time-to-solution by enabling orders-of-magnitude more objective evaluations within a fixed time budget.
- **Foundation for digital twin integration:** This work provides a physics-consistent thermal core that can be coupled with online sensing and higher-level decision layers in a digital twin. The low computational footprint supports deployment on edge/near-edge hardware for monitoring and prediction, while the continuous field representation supports downstream coupling to microstructure or distortion models and supports data assimilation workflows.

Overall, the proposed PINN framework is not intended to replace high-fidelity FEM/FVM for all analyses; rather, its value is as a fast, physics-consistent, and differentiable surrogate that enables workflows, real-time control, high-throughput optimization, and digital-twin deployment, that are difficult or infeasible with conventional solvers alone.

An important observation for interpretation is that the present formulation solves a conduction-dominated thermal model with an effective moving volumetric heat source. Therefore, while it is well suited for predicting temperature histories and derived thermal-cycle metrics (e.g., peak temperature, cooling rate, etc), it does not explicitly resolve weld-pool fluid flow, latent heat effects, or keyhole dynamics. As a consequence, applications that require accurate prediction of fusion-zone geometry or deep penetration across wide thickness ranges may require (i) an enriched physics model (e.g., enthalpy-based phase change and/or transport effects) and/or (ii) a more expressive heat-source representation tailored to through-thickness power deposition. These extensions are natural directions for future work and should be considered when deploying the framework beyond the validated regime.

7. Conclusion

This study addressed the persistent challenge of creating a computationally efficient yet physically accurate method for simulating the 3D thermal dynamics of MIG welding. A novel framework based on Physics-Informed Neural Networks was successfully developed and validated. The framework accurately predicts transient thermal histories, achieving l_2 errors as low as 2.28 K when validated against experimental data. This high level of accuracy is achieved with a computational time of approximately 14 min, representing a significant speedup over conventional simulation methods. A systematic hyperparameter study provided insightful information about the convergence, stability, and fidelity of the MIG temperature predictions. This work demonstrates the potential of PINNs to serve as a powerful tool in digital manufacturing, enabling rapid process design, optimization, and providing a foundation for the development of real-time digital twins for complex manufacturing processes like welding.

A natural extension of the present work is to enrich the PINN with additional physics by incorporating temperature-dependent properties and phase change directly in the PDE residual, e.g., via an enthalpy formulation. This should improve accuracy in high-temperature regions and near the fusion boundary. The framework is well suited to inverse problems: using sparse temperature data (thermocouples, pyrometry) to infer uncertain parameters such as heat-source efficiency, absorptivity, or convective coefficients. Treating these as trainable parameters enables joint state-parameter estimation, reducing experimental burden while improving fidelity.

Finally, coupling the transient thermal model with solid mechanics (thermal strains and, where appropriate, elastoplasticity) would allow prediction of residual stresses, distortions, and dimensional tolerances. Such a thermo-mechanical extension would provide a unified, efficient tool linking pool morphology to final stress state and weld quality.

CRediT authorship contribution statement

Carlos Antonio Ribeiro Duarte: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Funding acquisition, Formal analysis, Conceptualization; **André Alves de Resende:** Writing – review & editing, Visualization, Investigation, Funding acquisition; **Francisco José de Souza:** Writing - review & editing, Resources, Methodology; **João Rodrigo Andrade:** Writing – review & editing, Methodology, Formal analysis; **Luiz Eduardo dos Santos Paes:** Writing – review & editing, Visualization, Methodology, Investigation; **Leonardo Rosa Ribeiro da Silva:** Writing – review & editing, Project administration, Formal analysis; **João Marcos Souza Dias:** Writing – review & editing, Formal analysis, Data curation; **Rafael de Oliveira Teloli:** Writing – review & editing, Resources, Investigation, Formal analysis; **Bruno Jacob:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] F. Sartori, R.H. G.e. Silva, J.C. Dutra, P.L.E. dos Santos, M.B. Schwedersky, C. Marques, Uma análise comparativa entre diferentes versões de variantes modernas do processo MIG/MAG para o passe de raiz em soldagem orbital, Soldagem Inspecao 22 (4) (2017) 442–452. https://doi.org/10.1590/0104-9224/SI2204.04

[2] T.R. de Castro, L.E. dos Santos Paes, J.M.S. Dias, A.G.M. Santos, T.M.D. Borba, J.R. Andrade, S.D. Franco, E. dos Santos Magalhães, L.O. Vilarinho, GMAW root pass of shipbuilding steel plates with different thicknesses, Int. J. Adv. Manuf. Technol. 134 (1) (2024) 171–189. https://doi.org/10.1007/s00170-024-14142-1

[3] A.A. de Resende, F. Keocheguerians, L.O. Vilarinho, The influence of CO2 and O2 content on globular spray transition current when using argon-based blends in GMAW of ER 70s 6 wire, Weld. Int. 24 (8) (2010) 593–601. https://doi.org/10.1080/09507110903568828

[4] R.H. Gonçalves e Silva, L.E. dos Santos Paes, G.L. de Sousa, C. Marques, A.B. Viviani, M.B. Schwedersky, T.L.F. da Costa Pinto, Design of a wire measurement system for dynamic feeding TIG welding using instantaneous angular speed, Int. J. Adv. Manuf. Technol. 101 (5) (2019) 1651–1660. https://doi.org/10.1007/s00170-018-3026-2

[5] F.C. Neto, M.C. Fredel, M. Pereira, P.L.E. dos Santos, Laser power modulation to grain refinement of SAE 1045 steel welds, J. Laser Appl. 32 (2) (2020) 022027. https://doi.org/10.2351/7.0000096

[6] R.G. Dourado da Silva, E. dos Santos Magalhães, S.M.M. de Lima e Silva, L.E. dos Santos Paes, M. Pereira, Estimating the absorption efficiency in a laser welding process using a nonlinear inverse problem method, Int. J. Therm. Sci. 183 (2023) 107846. https://doi.org/10.1016/j.ijthermalsci.2022.107846

[7] L.E. dos Santos Paes, J.R. Andrade, F.S. Lobato, E. dos Santos Magalhães, V. Ponomarov, F.J. de Souza, L.O. Vilarinho, Sensitivity analysis and multi-objective optimization of tungsten inert gas (TIG) welding based on numerical simulation, Int. J. Adv. Manuf. Technol. 122 (2) (2022) 783–797. https://doi.org/10.1007/s00170-022-09934-2

[8] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, J. Comput. Phys. 378 (2019) 686–707.

[9] X. Meng, G.E. Karniadakis, A composite neural network that learns from multi-fidelity data: application to function approximation and inverse PDE problems, J. Comput. Phys. 401 (2020) 109020.

[10] M.J.H. Rawa, M.H. Razavi Dehkordi, M.J. Kholoud, N.H. Abu-Hamdeh, H. Azimy, Using the numerical simulation and artificial neural network (ANN) to evaluate temperature distribution in pulsed laser welding of different alloys, Eng. Appl. Artif. Intell. 126 (2023) 107025. https://doi.org/10.1016/j.engappai.2023.107025

[11] M. Jarwitz, A. Michalowski, Application of output constraints to a physics-informed hybrid model for the prediction of the threshold of deep-penetration laser welding, Procedia CIRP 124 (2024) 789–792. 13th CIRP Conference on Photonic Technologies [LANE 2024], 15–19 September 2024, Fürth, Germany, https://doi.org/10.1016/j.procir.2024.08.226

[12] E. King, Y. Li, S. Hu, E. Machorro, Physics-informed machine-learning model of temperature evolution under solid phase processes, Comput. Mech. 72 (1) (2023) 125–136. https://doi.org/10.1007/s00466-023-02289-9

[13] P. Scheel, E. Hosseini, Pulse approach: a physics-guided machine learning model for thermal analysis in laser-based powder bed fusion of metals, Progress Additive Manuf. (2024). https://doi.org/10.1007/s40964-024-00713-x

[14] M. Shafaie, M. Sarparast, H. Zhang, Integrating data-driven system to predict temperature and distortion in multi-layer direct metal deposition processes, Int. J. Adv. Manuf. Technol. 134 (1) (2024) 545–555. https://doi.org/10.1007/s00170-024-14082-w

[15] S. Li, G. Wang, Y. Di, L. Wang, H. Wang, Q. Zhou, A physics-informed neural network framework to predict 3D temperature field without labeled data in process of laser metal deposition, Eng. Appl. Artif. Intell. 120 (2023) 105908. https://doi.org/10.1016/j.engappai.2023.105908

[16] S. Guo, M. Agarwal, C. Cooper, Q. Tian, R.X. Gao, W. Guo, Y.B. Guo, Machine learning for metal additive manufacturing: towards a physics-informed data-driven paradigm, J. Manuf. Syst. 62 (2022) 145–163. https://doi.org/10.1016/j.jmsy.2021.11.003

[17] E. Hosseini, P. Scheel, O. Müller, R. Molinaro, S. Mishra, Single-track thermal analysis of laser powder bed fusion process: parametric solution through physics-informed neural networks, Comput. Methods Appl. Mech. Eng. 410 (2023) 116019. https://doi.org/10.1016/j.cma.2023.116019

[18] S. Hartmann, O. Murua, J.I. Arrizubieta, A. Lamikiz, P. Mayr, Physics-informed neural network approach to speed up laser-DED modelling, Procedia CIRP 124 (2024) 352–357. 13th CIRP Conference on Photonic Technologies [LANE 2024], 15–19 September 2024, Fürth, Germany, https://doi.org/10.1016/j.procir.2024.08.132

[19] M. Ryan, M.H. Baqershahi, H. Moshayedi, E. Ghafoori, Physics-informed machine learning surrogate for scalable simulation of thermal histories during wire-arc directed energy deposition, Additive Manuf. Lett. 15 (2025) 100327. https://doi.org/10.1016/j.addlet.2025.100327

[20] A. Fernández-Zabalza, F. Veiga, A. Suárez, V. Uralde, X. Sandua, J.R. Alfaro, Application of symmetric neural networks for bead geometry determination in wire and arc additive manufacturing (WAAM), Symmetry 17 (3) (2025). https://doi.org/10.3390/sym17030326

[21] A.A. de Resende, C.A.R. Duarte, The role of sustainability in the welding process: context, technologies and challenges, Environ. Dev. Sustain. (2025). https://doi.org/10.1007/s10668-025-06976-w

[22] S. Baruah, I.V. Singh, A framework based on nonlinear FE simulations and artificial neural networks for estimating the thermal profile in arc welding, Finite Elem. Anal. Des. 226 (2023) 104024. https://doi.org/10.1016/j.finel.2023.104024

[23] L.D. McClenny, U.M. Braga-Neto, Self-adaptive physics-informed neural networks, J. Comput. Phys. 474 (2023) 111722.

[24] S.J. Anagnostopoulos, J.D. Toscano, N. Stergiopulos, G.E. Karniadakis, Residual-based attention in physics-informed neural networks, Comput. Methods Appl. Mech. Eng. 421 (2024) 116805.

[25] W. Chen, A.A. Howard, P. Stinis, Self-adaptive weights based on balanced residual decay rate for physics-informed neural networks and deep operator networks, arXiv preprint arXiv:2407.01613 (2024).

[26] D.P. Kingma, Adam: a method for stochastic optimization, arXiv preprint arXiv:1412.6980 (2014).

[27] A.G. Baydin, B.A. Pearlmutter, A.A. Radul, J.M. Siskind, Automatic differentiation in machine learning: a survey, 2018, https://arxiv.org/abs/1502.05767. 1502.05767

[28] O. Champagne, X.-T. Pham, Numerical simulation of moving heat source in arc welding using the element-free Galerkin method with experimental validation and numerical study, Int. J. Heat Mass Transf. 154 (2020) 119633. https://doi.org/10.1016/j.ijheatmasstransfer.2020.119633

[29] L.E. dos Santos Paes, J.R. Andrade, M.G. Prates, D.D. B.G. de Souza, S.L. Brião, F.S. Lobato, E. dos Santos Magalhães, B.T.P. Jacob, R.P. Reis, L.O. Vilarinho, Path planning strategies for hardness improvement employing surface remelting in AISI 1045 steel, Surf. Coat. Technol. 425 (2021) 127728. https://doi.org/10.1016/j.surfcoat.2021.127728

[30] L.E. dos Santos Paes, J.M.S. Dias, J.R. Andrade, E.B. Filho, H.N. Ferraresi, L.R. Ribeiro da Silva, C. Xavier de Jesus Silva, V.L. Borges, K.C. Riffel, S. Hereñú, P. Francia, E. dos Santos Magalhães, M.L. Lagares Junior, C.A. Ribeiro Duarte, T. Vieira da Cunha, N. dos Santos Saad, L.O. Vilarinho, Comprehensive experimental and numerical characterization of microstructural and mechanical anisotropy in wire arc additive manufactured carbon steel, J. Mater. Res. Technol. 36 (2025) 7244–7260. https://doi.org/10.1016/j.jmrt.2025.04.318

[31] D.P. Kingma, J. Ba, Adam: a method for stochastic optimization, 2017, https://arxiv.org/abs/1412.6980. 1412.6980

[32] R. Jin, W. Chen, A. Sudjianto, An efficient algorithm for constructing optimal design of computer experiments, J. Stat. Plan Inference 134 (1) (2005) 268–287. https://doi.org/10.1016/j.jspi.2004.02.014